The **FACTOR CLASSES** are variables selected by the user and provide important information about *environmental conditions, soil attributes*, *analytical methods*, and are used to calculate the **SCORES**.

In summary:

ENVIROMENTAL FACTORS

- **Region Class** indicates if the region is umid or arid.
- **Season Class** indicates the season of soil sampling.
- **Climate Class** indicates the combination of sunny days and precipitation of sampling site.
- **Slope Class** indicates the slope of the sampling site.

SOIL FACTORS

- **Texture Class** indicates soil texture (granulometric distribution).
- **Mineral Class** indicates which type of clay mineral is dominant in the clay fraction.
- **Organic Matter Class** indicates the level of soil organic matter.
- **Iron Oxide Class** indicates if the soil is Ultisol or not.
- **Weathering Class** indicates the weathering intensity based on the presence of carbonates and soil classification.

METHOD FACTORS

- **P Method Class** indicates which method was used in assessing P availability.
- **EC Method Class** indicates which method was used to measure electric conductivity.

ENVIROMENTAL FACTORS

Region Class

(1) Arid region (annual precipitation ≤ 25mm).
(2) Umid region (annual precipitation > 25mm).

Season Class

(1) Sampling in spring, before harvest or during sowing.
(2) Sampling in summer, during plant growth.
(3) Sampling in autumn, during harvest or just after.
(4) Sampling in winter, during fallow or cover crop.

Climate Class

(1) More than 170 sunny days and more than 550mm annual precipitation (high / high).
(2) More than 170 sunny days and less than 550mm annual precipitation (high / low).
(3) Less than 170 sunny days and more than 550mm annual precipitation (low / high).
(4) Less than sunny days and less than 550mm annual precipitation (low / low).

Slope Class

(1) Slope between 0 and 2%.
(2) Slope between 2 and 5%.
(3) Slope between 5 and 9%.
(4) Slope between 9 and 15%.
(5) Slope over 15%.

SOIL FACTORS

Texture Class

(1) Sand, loamy sand, or sandy loam (with < 8% clay).
(2) Sandy loam (with >8% clay), sandy clay loam, or loam.
(3) Silt loam, silt.
(4) Sandy clay, clay loam, silty clay loam, silty clay, or clay (with <60% clay content)
(5) Soils with clay content over 60%.

Mineral Class

(1) Clay fraction with predominance of smectites.
(2) Clay fraction with predominance of low crystallinity clays (e.g., allophane, imogolite).
(3) Clay fraction with predominance of other types of clays (e.g., kaolinite).

Organic Matter Class *

* Soils from WRB or other classification system must be classified using Soil Taxonomy or SIBCS

(1) Very high organic matter content
- **Soil Taxonomy:** All Histosols; Andisols, Oxisols and Spodosols with aquic regime; Turbels e Histels.
- **SIBCS:** *Organossolos Tiomórficos, Fólicos, Háplicos; Cambissolos Hísticos*.

(2) High organic matter content
- **Soil Taxonomy:** All Mollisols; Andisols of umid regions (ustic and udic regime); Inceptisols with aquic regime; Vertisols with ustic, xeric and aquic regime; Ultisols with aquic regime or high organic matter (Humults); Oxisols with udic regime.
- **SIBCS:** *Chernossolos Rêndzicos, Ebânicos, Argilúvicos, Háplicos; Cambissolos Húmicos; Gleissolos Melânicos, Tiomórficos; Latossolos Brunos; Nitossolos Brunos; Vertissolos Ebânicos, Hidromórficos; Argissolos Bruno-acinzentados*.

(3) Medium organic matter content
- **Soil Taxonomy:** All Alfisols; Andisols with cryic regime and Vitrands. Andisols and Gelisols that do not belong to organic matter classes (1), (2) and (4); Spodosols with cryic regime; Vertisols with cryic regime; Entisols with aquic regime; Inceptisols with plaggen or anthropic epipedon (Anthrepts).

- **SIBCS:** *Vertissolos Háplicos; Nitossolos Vermelhos, Háplicos; Latossolos Amarelos, Vermelhos, Vermelho-amarelos; Argissolos Acinzentados, Amarelos, Vermelhos, Vermelho-amarelos; Cambissolos Háplicos; Gleissolos Háplicos; Cambissolos Flúvicos*.

(4) Low organic matter content

- **Soil Taxonomy:** All Aridisols; Andisols with torric and xeric regime; Entisols with xeric regime, Arents and Orthents; Inceptisols and Spodosols that do not belong to organic matter classes (1), (2) and (3). Oxisols with ustic and torric regime, Orthox; Vertisols with torric regime; Ultisols with udic and ustic regime.
- **SIBCS:** *Plintossolos Háplicos, Pétricos, Argilúvicos; Neossolos Litólicos, Flúvicos, Quartzarênicos, Regolíticos; Gleissolos Sálicos; Planossolos Nátricos, Háplicos; Luvissolos Crômicos, Háplicos; Espodossolos Humilúvicos, Ferriúvicos, Ferri-humilúvicos*.

Iron Oxide Class

(1) **Soil Taxonomy:** Ultisols, soils with argillic or kandic horizon and low base saturation, with any water and temperature regime, except from aridic.
SIBCS: soils with textural change (*B textural*) that comprehend *Argissolos* and *Planossolos*.

(2) All other soils.

Weathering Class

(1) Soils with carbonates.

(2) Soils highly weathered (1:1 clays, oxides and/or hydroxides of Fe and Al).

(3) All other soils.

METHOD FACTORS

P Method Class

(1) Phosphorus availability assessed by Mehlich 1.
(2) Phosphorus availability assessed by Mehlich 3.
(3) Phosphorus availability assessed by Bray method.
(4) Phosphorus availability assessed by Olsen method.
(5) Phosphorus availability assessed by resin.
(6) Phosphorus availability assessed by iron oxide strip.

EC Method Class

(1) Electric conductivity measured by saturation paste.
(2) Electric conductivity measured using 1:1 ratio.